

Department of Clinical and Molecular Medicine

**Examination paper for
MOL3100 - Introduction to Molecular Medicine with project
MOL3000 - Introduction to Molecular Medicine**

Examination date: 11.12.21

Examination time (from-to): 9.00-13.00

Permitted examination support material: D (No printed or hand-written support material is allowed. A specific basic calculator is allowed.)

Academic contact during examination: Bodil Kavli, Phone: 97661442

OTHER INFORMATION

Make your own assumptions: If a question is unclear/vague, make your own assumptions and specify them in your answer. Only contact academic contact in case of errors or insufficiencies in the question set. Address an invigilator if you wish to contact the academic contact. Write down the question in advance.

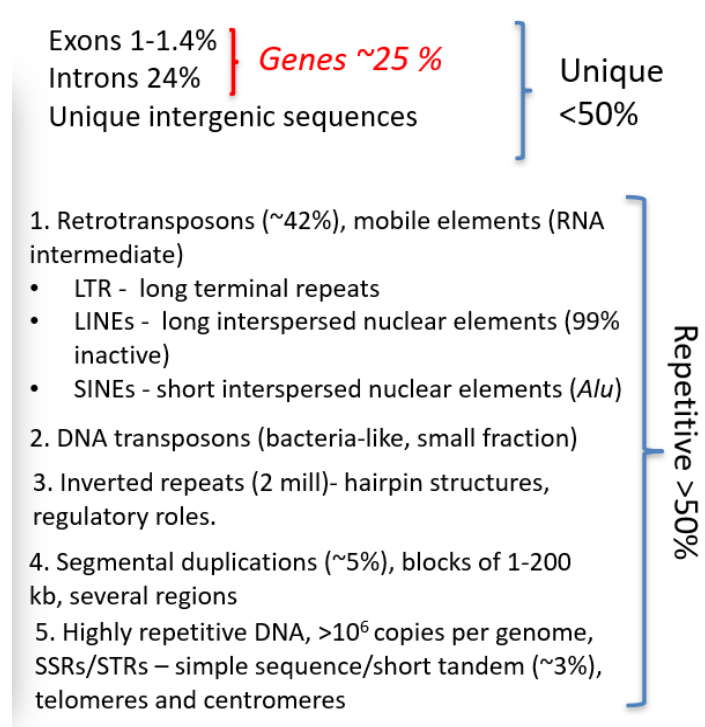
Notifications: If there is a need to send a message to the candidates during the exam (e.g., if there is an error in the question set), this will be done by sending a notification in Inspira. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Question 1 (6 p)

The human haploid genome contains 3.1 billion (10^9) base pairs organized in 23 nuclear chromosomes (+ mitochondrial DNA).

A. Describe the major composition elements in the human nuclear genome and indicate their relative prevalence (% of genome size) when known. (3p)



B. The human genome contains approximately 20 000 protein-coding genes, but more than 60 000 human genes (including pseudogenes) are currently annotated. Describe different classes of non-coding genes, indicate their gene products and some functional roles. (3 p)

The gene product of non-coding genes are various RNA molecules.

Ribosomal RNA, structural and catalytic role (ribosome)

tRNA (translation)

Long ncRNAs: > 200 nt

lncRNAs function in chromatin remodeling and regulation gene expression (at the level of transcription, epigenetic modification, translation)

Small ncRNAs: < 200 nt

miRNAs regulate gene expression post-transcriptionally (at mRNA level)

piRNAs silence transposons

snRNA splicing

snoRNA modification of rRNA, tRNA and snRNA

Question 2 (6 p)

Epigenetic modifications occur on both DNA and proteins.

A. What are the most common epigenetic modifications on the N-terminal histone tails (mention at least three different modifications) and indicate the amino acids modified (2 p)

- *Methylation; Lysine (K) (mono, di, tri), Arginine (R) (mono, di)*
- *Acetylation; Lys (K)*
- *Phosphorylation; (Serin (S), Threonine (T), Tyrosine (Y))*
- *PARylation (poly ADP ribose); (K)*
- *Ubiquitinylation; (K)*
- *SUMOylation (K)*

B. Histone modifications exert their effect via two main mechanisms. What are these mechanisms? (2 p)

1. *Directly influencing the overall structure of chromatin.*
2. *Regulating the binding of effector molecules (readers).*

C. At the gene level epigenetic modifications are dynamically regulated as part of a biological response. Describe the terms epigenetic writers, readers and erasers and give examples. (2p)

Writers: Adds modification on histone tails (or DNA), e.g. methyl transferases, acetyl-transferases, kinases, Ubi ligases, PARPs

Readers: Effector molecules that bind to modified histone tails. Eg Chromodomain of HP1 binds to H3K9me3

Erasers: Removes modification, e.g. demethylases, deacetylases, phosphatases, deubiquitinating enzymes.

Question 3 (4p)

Monogenic diseases can be autosomal, X-linked, Y-linked or mitochondrial (mtDNA). Autosomal and X-linked diseases can be dominant or recessive. Below is a family tree showing three generations of a family with a rare disease.

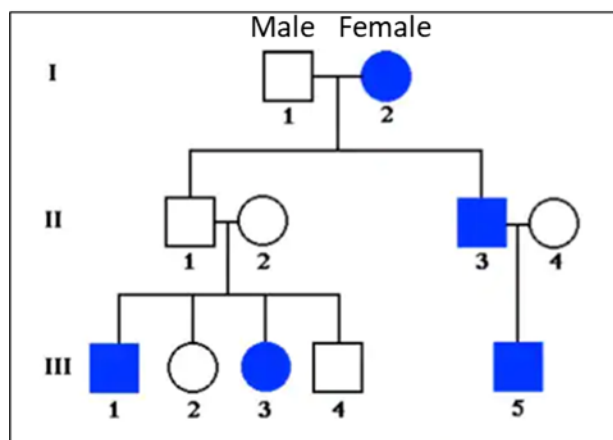


Figure explanation:

The figure illustrates 3 generations, marked I, II and III.

Parents and offspring are connected by lines between generations.

Persons with disease are indicated in blue.

What are the potential modes of inheritance of this disease? Range them from least likely to most likely and argue for your answer. (4p)

1. It is not Y-linked because females are affected
2. X-linked dominant is not likely because father to son transmission can be observed for person 5 in generation III. Boys inherit their X-chromosome from their mother, and the mother is healthy.
3. It is not likely to be mitochondrial caused by mtDNA mutation – if mitochondrial, persons 2 and 4 in generation II have to be asymptomatic mtDNA mutated mothers since mtDNA are maternally inherited. This is not likely since mitochondrial diseases are rare. Persons 2 and 4 in generation II are not related to individuals in generation I (and most probably not to each other).
4. X-linked recessive is not likely – if X-linked recessive person 4 in generation II has to be a carrier for the mutation to have an affected son. This is a rare disease. In addition, person 1 in generation II would be expected to have the disease since his mother is homozygous and he will inherit his X-chromosome from her.
5. Autosomal recessive is not very likely. Because person 3 in generation II has the disease, this would require that the father in generation I is a carrier, and also that person 2 in generation II is a carrier – and that is not very likely since this is a rare disease in the population.
6. Autosomal dominant is most likely. Both sexes are affected, and the disease occurs in (almost) every generation. The skipping in generation II (for person 1) is most likely due to reduced penetrance (and less likely due to new mutation in the germ cells of person 1 or person 2 in generation II)

Question 4 (2 p)

Nucleotide substitutions account for ~70% of the identified disease-causing mutations in humans. About 50% are missense mutations, whereas only 10% are nonsense mutations. In general, nonsense mutations are more severe than missense mutations. So why are nonsense mutations not seen more frequently in human disease? Argue for two mechanism that may be involved.

1. *A nonsense mutation involves the formation of a new stop codon (TAA, TAG, TGA). Thus, there are a limited number of amino acid codons that will generate a stop codon when mutated.*
2. *A nonsense mutation is often more deleterious than a missense mutation because it generates a truncated polypeptide (and often a total loss of function). Such a mutation may be more efficiently selected against by purifying selection.*

Question 5 (2 p)

Familial hypercholesterolemia (FH) is the most common autosomal dominant disorder, with a prevalence of 1 in 350-500 in most populations. Cystic fibrosis (CF) is the most common autosomal recessive disorder in Caucasians, with a prevalence at birth of about 1 in 3000. For both diseases, most of the disease-causing mutations are “loss-of-function” mutations. Explain why FH is dominant, and CS recessive?

FH is dominant because of haploinsufficiency – that is, the remaining normal allele cannot fully compensate for the loss of the other allele. However, loss of both alleles creates a more severe phenotype.

CS is recessive because the remaining normal allele is sufficient to compensate for the loss of the other allele. Loss of both alleles are required to express the phenotype.

Question 6 (2 p)

A new treatment option for CF, the triple combination therapy “Trikafta”, was approved by the U.S Food and Drug Administration in 2019 for use in patients with the most frequent CF mutation, the Phe508del mutation in exon 10 of the *CFTR* gene.

The *CFTR* gene has 27 exons and encode 1480 amino acids. Gly542X, a nonsense mutation in exon 11 of the *CFTR* gene, is the second most frequent CF mutation in several European countries. Would “Trikafta” be effective against the Gly542X mutation? Argue for your answer.

The 542X mutation produces a stop codon – thus, no protein is produced. Therefore, Trikafta would not work on this mutation.

Question 7 (4p)

Receptor proteins are central components in biological communication through signal transduction. Activation of signalling proteins (gain of function mutation of proto-oncogenes) are important in cancer.

A. We have multiple receptor types on/in our cells that can be classified according to cellular location. Mention at least 3 types of transmembrane receptors and one intracellular receptor. (2p)

G-protein coupled receptors, receptor tyrosine kinases, cytokine receptors, Ion-channels and androgen receptors (intracellular)

B. Describe the main players in a signal transduction pathway including a transmembrane receptor, RAS (Rat sarcoma virus) GTPase, and the MAPK (mitogen-activated protein kinase) cascade. (2p)

EGFR (dimer, RTK), RAS- (monomeric G-prot.) linked to RTK and activated via GRB2-Sos, activated Ras recruited and activate RAF (MAPKKK), which activate MEK1/2 (MAPKK), which activate ERK1/2 (MAPK)

Question 8 (4 p)

Activation of signalling proteins (gain of function mutation of proto-oncogenes) are important in cancer.

A. EGFR (epidermal growth factor receptor) signalling can activate multiple downstream pathways. Mention at least two pathways (2p)

RAS-MAPK, PI3K-mTOR, PLC-γ-JAK-STAT

B. Describe at least four possible ways of inhibiting EGFR signalling by targeted treatment? (2p)

mAbs to EGFR, EGFR inhibitors, B-RaFi, MEKi, JAKi, PI3Ki and mTOR inhibitors

Question 9 (4 p)

Immune checkpoint inhibitors have revolutionized cancer therapy. However, severe limitations are reported. Describe the main concept of this new treatment (mechanism of action) and list at least three important limitations to checkpoint immune therapies. (4p)

The main concept is to target immune checkpoints, key regulators of the immune system that dampen the immune response to an immunologic stimulus. Some cancers can protect themselves from attack by stimulating immune checkpoint targets. Checkpoint therapy blocks inhibitory checkpoints, restoring immune system function to attack the cancer cells (removes the break). Currently approved checkpoint inhibitors target the receptors CTLA4 or PD-1, or the PD-L1 ligand.

Limitations: Immune related toxicity /autoimmune diseases, acquired resistance due to compensatory expression of other checkpoint receptors, changed MHC expression, low levels of neoantigens, cost, change in TME and signalling

Question 10 (8p)

Bioinformatics is an interdisciplinary field using informatics to explore biological issues. An important task in bioinformatics is to find protein sequences that are similar. This is done by creating algorithms for sequence alignment.

- A. What is meant by global and local alignments? (2p)
- B. What are the main advantages of using local alignments? (2p)
- C. Describe how BLAST use local alignments to limit the number of final alignments in a database search. (2p)
- D. Describe how the local alignments are used further to perform a banded search. (2p)

A. Describe what is meant by global and local alignments

Answer: In a global alignment we compare the full length of the first sequence against the full length of the second sequence, looking for similarities across the full length of the two sequences. In a local alignment we just focus on a sub-region of each sequence where there is a significant similarity between the two sequences and ignore the rest.

B. What are the main advantages of using local alignments?

Answer: Local alignments are better for comparing sequences with gaps caused by insertions and deletions. Local alignments can be broken down to subtasks during dynamic programming.

C. Describe how BLAST (Basic Local Alignment Search Tool) use local alignments to limit the number of final alignments in a database search?

Answer:

Step 1 – Split the query sequence into small subsequences of length k (“words”), for example k=3 is common.

Step 2 – Use the scoring matrix to find the words with highest conservation (highest score).

Step 3 – Each database sequence is compared against the most high-scoring words (speeds up process).

Step 4 - Only sequences with many matches to high-scoring words are used for final alignments.

D. Describe how the local alignments are used further to perform a banded search.

Answer: Look for high scoring matches along the diagonal of the dot matrix. Expand around the diagonal. Ignore other parts of the dot (similarity) matrix.

MCQ (8p, 1p each)

Only one letter (A,B,C or D) for each question (the best alternative).

1.

A bacterial strain may develop from fully susceptible to highly resistant to an antibiotic drug in one step or in several steps.	
Which mechanism is most likely to lead to development of resistance through several steps?	
A	Transformation
B	Transduction
C	Conjugation
D	Mutation

2.

Antimicrobial drugs use different mechanisms to inhibit bacterial growth.	
What is the target of beta-lactam antibiotics like penicillin?	
A	Cell wall synthesis
B	DNA synthesis
C	Protein synthesis
D	Folate synthesis

3.

Trans-differentiation is a strategy to develop patient-specific cells that can be used in personalized therapy.	
Trans-differentiation refers to a process in which:	
A	differentiated cells are pushed towards less differentiated state
B	differentiated cells are pushed towards more differentiated state
C	differentiated cells are pushed towards pluripotent state
D	differentiated cells are pushed towards lineage of interest

4.

Mutations are often involved in disease, but also required for evolution	
A mutation is defined as:	
A	a change in the DNA sequence of a gene that causes disease
B	a change in the DNA sequence that causes disease
C	a change in the DNA sequence away from the normal
D	a change in the DNA sequence of a gene away from normal

5.

Atherosclerosis is the major cause of myocardial infarction and stroke.	
Which of this statement about atherosclerosis is not correct?	
A	Plaque disruption leads to the formation of blood clots
B	Intimal thickening eventually blocks the entire blood vessel leading to stroke or myocardial infarction
C	Atherosclerosis is a gradual process starting from diffuse to pathological intimal thickening
D	Atherosclerosis can be caused by both genetic and environmental factors

6.

Endothelial cell dysfunction is a central part in the process of atherosclerosis.	
Which factors contribute to impaired endothelial cell function	
A	increased level of nitric oxide and increased level of oxygen radicals
B	increased level of nitric oxide and decreased level of oxygen radicals
C	decreased levels of nitric oxide and increased level of oxygen radicals
D	decreased level of nitric oxide and decreased level of oxygen radicals

7.

Diagnostic criteria for diabetes usually include fasting plasma levels of glucose, plasma glucose levels two hours after glucose load, and HbA1c in %.	
What is the molecular mechanism underlying HbA1c?	
A	Hyperglycemia in red and white blood cells
B	Molecular binding of glucose to hemoglobin
C	Glucose toxicity in red blood cells
D	Diabetes complication in red blood cells

8.

According to WHO classification, types of diabetes include type 1, type 2, hybrid forms, other specific types, unclassified and hyperglycemia during pregnancy.	
Which is the most common molecular mechanism underlying both type 1 and type 2 diabetes?	
A	Impaired insulin secretion
B	Impaired glucose transporter type 4
C	Insulin resistance
D	Immuno-inflammation in pancreatic islets